

Deep Learning-based Semantic Search Techniques for Enhancing Product Matching in E-commerce

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Abstract—Searching is the process of information retrieval utilizing specific criteria or keywords. Integrating search functionalities on e-commerce platforms enables users to efficiently locate exactly what they are searching for through keyword matching. Beyond conventional keyword matching, semantic search involves aligning products with customer queries by capturing the essence of the queries, thereby retrieving semantically related products from the pertinent catalog. Semantic search enhances the e-commerce shopping experience by allowing platforms to tailor responses to user preferences through an in-depth understanding of search intents. Challenges such as morphological variations, spelling errors, and the interpretation of synonyms, antonyms, and hypernyms are addressed through deep learning models designed for semantic query-product matching. This study conducts a comparative analysis of various semantic search methodologies and assesses their efficacy, incorporating deep learning strategies for query auto-completion and spelling corrections. The evaluation employs sentence transformer models to determine the optimal approach for semantic searching, gauged by nDCG, MRR, and MAP metrics. LSTM, BART, and n-gram models are also examined for auto-completion capabilities. The research analyzes the Amazon Shopping Queries Dataset and the Upstart Commerce catalog datasets.

Index Terms—Semantic product searching, Natural Language Processing (NLP), Machine Learning, E-commerce

I. INTRODUCTION

Text and search play a crucial role in today's digital age, and their importance can be seen in various aspects of our personal and professional lives, as well as in the functioning of modern society. This includes accessing and retrieving information from vast amounts of data. Online shopping and e-commerce heavily depend on search functions. At present, a product search engine works by taking a customer query as

an input that gets passed through a lexical matching engine, which then produces a match set (all products that contain the words present in the query) [1].

Searching involves discovering information by applying specific criteria or keywords. It's a simple process where you systematically look for and retrieve the data you need. Semantic searching goes beyond traditional keyword-based searches, focusing on the meaning behind user queries to deliver more contextually relevant results. It leverages natural language understanding and semantic analysis to enhance search accuracy, providing users with information that aligns better with their intent.

In the context of e-commerce, Semantic Product Search (SPS) offers potential benefits for an enhanced shopping experience. However, several challenges must be addressed for effective implementation. This research aims to assess the performance of semantic search algorithms in accurately interpreting user queries and providing relevant product results. It also examines the impact of allowing descriptive queries on user interactions and evaluates the integration of AI-driven technologies in enhancing the shopping journey. The study further scrutinizes the system's ability to understand user intent and deliver personalized results.

This research paper compares several models for semantic searching which are then trained on the open-source Amazon dataset containing query-product pairs that have been cleaned and pre-processed to extract queries relevant to the product catalog of UpStart Commerce. When conducting exploratory data analysis on this dataset, cosine similarity was leveraged to determine the most suitable matches between product titles and categories. Five pretrained sentence transformer models were employed to handle the task of semantic product searching and three approaches for sentence autocompletion were as-

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sessed. These models were then evaluated using standardized evaluation metrics to quantitatively analyze and compare their performance to choose the best performing model for further finetuning.

The rest of the paper is organized as follows. In Section 2 of this paper, an overview of relevant literature is presented. Section 3 contains a thorough description of the dataset utilized for this research. Subsequently, section 4 introduces the approaches for the semantic product searching and auto-completion tasks, while section 5 details the results of the experimentation, further elaborating on the dataset division, the experimental set, and the reasoning for the selected evaluation criterion. This is followed by section 6 comprising the conclusion of the research and acknowledgments in section 7.

II. RELATED WORK

In the field of semantic search, the effort to better match user queries with digital content has driven an evolution in methods and technologies. This literature review explores some relevant works that have shaped the current landscape of semantic search.

Reddy et al. [2] introduced the Shopping Queries Dataset as a significant benchmark for semantic matching research, focusing on tasks like Query-Product Ranking and Multiclass Product Classification. They employed the MS MACRO Cross-Encoder and a Multilayer Perceptron classifier, revealing the potential of neural approaches for improving semantic search, especially in multilingual contexts.

Nigam et al. [1] tackled the intricacies of semantic matching in product search by proposing a model that refines lexical matches with deep learning insights from customer behavior data. Their approach underscores the synergy between traditional search methods and modern deep learning, pointing to significant improvements and highlighting the ongoing challenges in precision and filtering.

Reimers and Gurevych [3] enhanced BERT with Sentence-BERT (SBERT), utilizing siamese and triplet networks for generating semantically meaningful sentence embeddings more efficiently. This approach addressed the computational and contextual limitations of existing models, setting a new standard for semantic relevance and search efficiency.

Jin et al. [4] explored product attribute value extraction, presenting a model that integrates a label encoder and semantic matching to reduce the need for human annotation. This work underscores the importance of understanding semantic nuances in product information, significantly advancing multi-label text classification tasks.

In a novel approach to semantic indexing, Jin et al. [5] developed LMINDEXER, leveraging generative language models to create neural sequential discrete representations. This method offers a promising direction for bypassing traditional text embedding challenges, highlighting the potential for knowledge graphs and ontologies in enriching semantic search.

Ravfogel et al. [6] critiqued existing information retrieval methods for abstract queries, presenting a model that utilizes

GPT3 for generating descriptions and a dual-encoder for sentence embedding. Their work provides valuable insights into handling vague descriptions, though it also notes the limitations of generative approaches in semantic clarity.

Choi et al. [7] proposed a methodology that combines transformers with user click data to enhance document representation and query matching. This approach, focusing on structured matching and the integration of lexical features, indicates a path toward more robust models for field-specific queries.

Muhammed et al. [8] introduced a four-tier training program using a bi-encoder architecture to optimize BERT-like models for product search with reduced latency. This model's development through pre-training and fine-tuning stages showcases the potential for incorporating broader product dimensions in search algorithms.

By employing a cross-encoder BERT model, Bagheri Garakani et al. [9] tackled semantic search through ranking applications based on offline generated scores. Their methodology highlights the effectiveness of re-ranking features and relevance estimators in real-time product search scenarios.

Chang et al. [10] adopted a tree-based Extreme Multi Label Classification (XMC) strategy, utilizing the XR-Linear PECOS model to enhance semantic product search matching. Their work demonstrates the complementarity of XMC approaches to existing lexical-based methods, offering improved performance with low latency.

Mandal et al. [11] proposed a method for refining semantic search results by training models to understand and map queries with similar meanings. Despite challenges in applying this offline-trained model, their approach emphasizes the importance of capturing the underlying semantics in query matching.

Lakshman et al. [12] focused on dyadic neural embedding models for large-scale semantic product searching, illustrating the potential of modeling dyadic data as a bipartite graph. Their work suggests future directions for end-to-end methods in graph partitioning and cluster prediction.

The foundational contributions of Devlin et al. [13] with BERT, and subsequent developments like DistilBERT by Sanh et al. [14] and MiniLM by Wang et al. [15], highlight the transformative impact of deep learning models on semantic search. These models enhance the understanding of context and enable fine-tuning for specific tasks, paving the way for more efficient and flexible search algorithms.

Choudhary et al. [16] introduced an attention-based method for extracting relevant information from large catalogs, employing hyperbolic entity modeling to improve search quality. This innovative approach underscores the significance of modeling relationships within search queries to enhance search accuracy.

Magnani et al. [17] combined traditional index-based and embedding-based neural retrieval approaches, illustrating the benefits of a two-tower structure in handling unique user queries. Their work points to future improvements in filtering

product descriptions to maximize the benefits of semantic models.

Deshmukh & Sethi [18] explored background linking tasks using SBERT for semantic search, demonstrating the potential of contextual representations in enhancing retrieval performance. Their methodology offers insights into the application of semantic search in providing context and insight for query articles.

Finally, Muennighoff [19] introduced SGPT, a unique strategy utilizing GPT models for semantic search, highlighting the effectiveness of sentiment embeddings and the distinction between bi-encoders and cross-encoders in handling asymmetric search scenarios. This study marks a significant advancement in phrase embeddings and model performance.

III. DATASET

The two datasets which were used are Amazon Shopping Queries Dataset and Upstart Commerce's catalog. The subsequent sections provide detailed descriptions of the datasets utilized.

A. Amazon Shopping Queries Dataset

The dataset under consideration in this work is the Amazon Shopping Queries Data set. This is the same dataset which is utilized by Reddy et al. [2]. This dataset consists of a variety of customer queries which have been annotated manually. There are two variants of this dataset. The large variant having 130,653 unique queries and 2,621,738 judgements against query-item pairs. The reduced variant contains 48,300 unique queries and 1,118,011 judgements against query-item pairs. The data has been divided into training and testing subsets.

The dataset is multilingual, containing English, Spanish and Japanese. There are two parquet files containing all the raw relevant information required for training and testing this dataset. They are namely "shopping_queries_dataset_examples" and "shopping_queries_dataset_products". For the shopping queries products file it contains the features shown in Table 1:

TABLE I
CONTAINS SAMPLE FEATURES FROM SHOPPING QUERIES PRODUCTS

Attributes	Examples
product_id	B003O0MNGC
product_title	Delta BreezSignature VFB25ACH 80 CFM Exhaust...
product_description	None
product_bullet_point	Virtually silent at less than 0.3 sones...
product_brand	DELTA ELECTRONICS (AMERICAS) LTD.
product_color	White
product_locale	US

Below in Table 2 of some of the queries present in the dataset alongside their ESCI value for the [US] locale:

TABLE II
CONTAINS SAMPLE QUERIES FROM SHOPPING QUERIES PRODUCTS

query	ESCI_Label
pink clipboards foldable	E
time-life world library ireland	S
henry's awful mistake	E
david bellamy watercolor	E

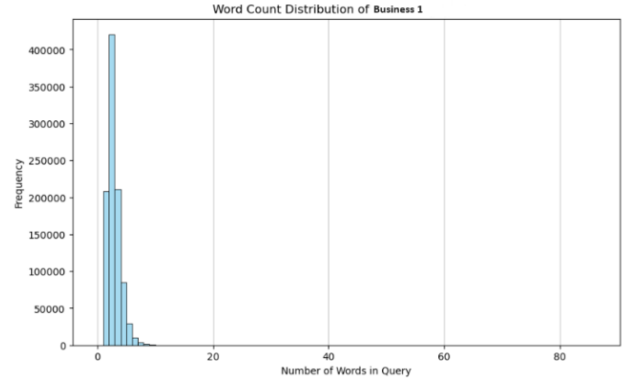


Fig. 1. Showing Word Count Distribution of Business1

B. UpStart Commerce Catalog

At present Upstart Commerce has its own catalog dataset. It is a comprehensive collection of product information. This dataset is organized at the SKU (Stock Keeping Unit) level, meaning that each row represents an individual variant of a product. It provides a rich and detailed source of information about a wide range of products from clients of Upstart Commerce.

Two distributions (Fig. 1 and 2) of search query lengths from Upstart Commerce's customer businesses websites to better elaborate the average amount of words per query have been added. For the sake of customer privacy, they will be anonymised and referred to as Business1 and Business2 respectively.

Remarks: Both the figures show clearly that most queries are around 2 to 6 words in length and nearly no queries make

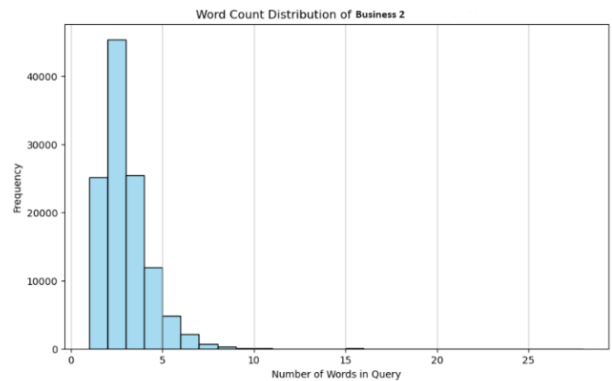


Fig. 2. Showing Word Count Distribution of Business2

it past the 10 word mark. This maps closely with the data trend for queries visible in the Amazon Shopping Queries Dataset as well.

C. Exploratory Data Analysis

The Amazon Shopping Queries dataset (this point on called Amazon data/dataset) was mapped to Upstart Commerce's catalog by creating a category column and affixing the product titles and descriptions. Embeddings of items in both datasets were compared which helped identify a suitable method for matching items from the Upstart Commerce catalog to queries available in the amazon data. The methods tried were: Matching product title column (Amazon) with category column (Catalog) directly. Matching product title (Amazon) with product title (Catalog) and appending the category associated with catalog title to the amazon products. A similar process was implemented for both methods, Japanese and EU locale entries were removed focusing on English products. U the paraphrase-MiniLM-L6-v2 model, embeddings were obtained. From metrics tried cosine similarity was chosen for matching. Special characters, HTML tags and emoticons were removed in data cleaning. CSVs were created based on cosine similarity thresholds, with focus on "title_055.csv" that has approximately 23,221 rows. Catalog data was processed to retain essential columns. The data frames were merged based on cosine similarity, associating categories with product names. The refined Amazon dataset, containing all columns paired with matched titles and categories, resulted from an inner join with the updated data frame.

IV. METHODOLOGY

A. Preprocessing of the Datasets

The Amazon dataset mentioned prior, has been sifted to extract only parts which were relevant to our collaborative company Upstart Commerce's requirements. The large variant mentioned prior, having 30,653 unique queries and 2,621,738 judgements against query-item pairs was utilized for this work. This dataset is multilingual, containing English, Spanish and Japanese. In accordance with the requirements and constraints, this work only takes the English portion of the dataset into consideration. That leaves originally 74.5% of the large dataset for that is the percentage of English queries.

B. Models for semantic searching

In this study, we conducted experiments utilizing five pre-trained models to facilitate semantic product search:

- 1) all-MiniLM-L6-v2: This is a sentence-transformer model which is used for clustering and semantic searching. Sentence embedding models are trained on very large datasets using a self-supervised contrastive learning objective for this model. The pretrained nreimers/MiniLM-L6-H384-uncased model is used and fine-tuned on a 1B sentence pairs dataset. Given an input text, it outputs a vector which captures the semantic information. The sentence vector may be used for information retrieval, clustering or sentence similarity tasks.

Here's the result of a sample query from the filtered amazon dataset that we are using. The query is matched with the product titles and descriptions and the top 10 results are displayed using the torch.topk function on the basis of cosine scores.

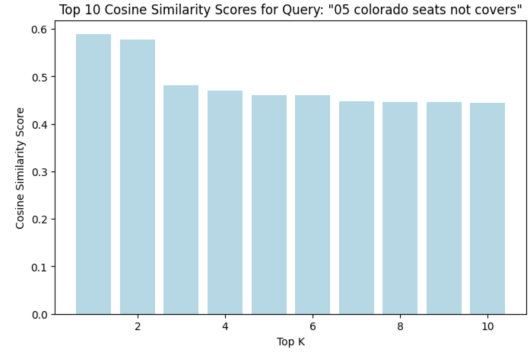


Fig. 3. Top 10 results of a sample query

Following are the percentages of E, S, C, I labels for this particular query

TABLE III
PERCENTAGES OF E, S, C, I LABELS USING 'ALL-MINI-L6-v2'

Labels	Percentages
E	70
S	10
C	0
I	10

- 2) msmarco-distilroberta-base-v2: MS MARCO is a large scale information retrieval corpus that was created based on real user search queries using Bing search engine. It concatenates the sentence embeddings $\mathbf{u}$ and $\mathbf{v}$ with the element-wise difference $|\mathbf{u} - \mathbf{v}|$ and multiply it with the trainable weight $\mathbf{W}_t \in \mathbb{R}^{3n \times k}$ [3]:

$$\mathbf{o} = \text{softmax}(\mathbf{W}_t(\mathbf{u}, \mathbf{v}, |\mathbf{u} - \mathbf{v}|)) \quad (1)$$

The training data consists of over 500k examples, while the complete corpus consists of over 8.8 Million passages. This model plays a significant role in advancing the field of semantic understanding and contributes to the development of more efficient and effective language models. The best 10 cosine similarity scores for this search that tells us how similar they are, is displayed here. We also use labels like Exact, Substitute, Complement, or Irrelevant to describe how well the product matches the search. We compare the search words with the product titles and descriptions, and the top 10 results are chosen based on how much they match.

Following are the percentages of E, S, C, I labels for this particular query

- 3) Msmarco-distilbert-base-v3: Msmarco-distilbert-base-v3 is a powerful pre-trained language model designed to excel in semantic understanding and natural language

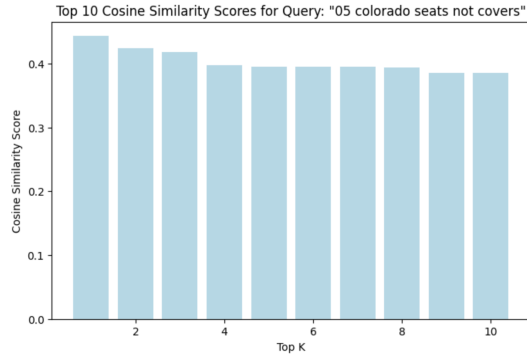


Fig. 4. Top 10 results of a sample query

TABLE IV
PERCENTAGES OF E, S,C, I LABELS USING
'MSMARCO-DISTILROBERTA-BASE-V2'

Labels	Percentages
E	30
S	10
C	40
I	20

processing tasks. Falling under the category of distil-BERT models, it represents a distilled or more compact version of the original BERT architecture. The top 10 cosine similarity scores whether they are Exact matches, Substitutes, Complements, or Irrelevant are shown below. The matching process involves comparing the query against the product titles and descriptions. The selection of the top 10 results is determined through the application of the torch.topk function, relying on the cosine scores for prioritization.

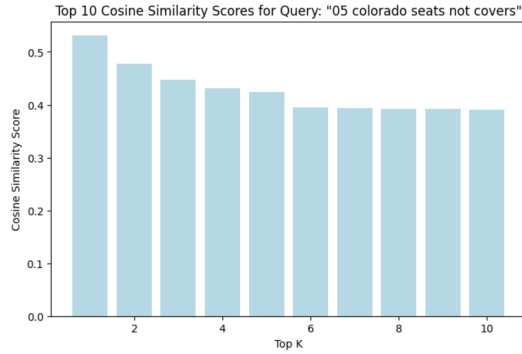


Fig. 5. Top 10 results of a sample query

Following are the percentages of E, S, C, I labels for this particular query

- 4) naver/efficient-splade-v-large-doc: Efficient SPLADE model is used for passage retrieval finding relevant passages of text from a given collection in response to a user query. Here are the top 10 results of cosine similarity score between product title, description and

TABLE V
PERCENTAGES OF E, S,C, I LABELS USING
'MSMARCO-DISTILBERT-BASE-V3'

Labels	Percentages
E	20
S	10
C	10
I	60

query. The descriptions are nan in some cases. The reason for this is that the description is missing in some of the columns of the dataset.

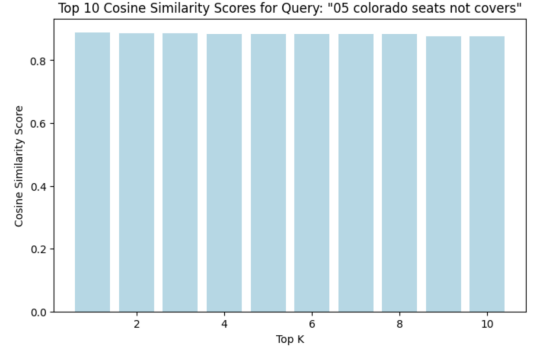


Fig. 6. Top 10 results of a sample query

Following are the percentages of E, S, C, I labels for this particular query

TABLE VI
PERCENTAGES OF E, S,C, I LABELS USING
'NAVER/EFFICIENT-SPLADE-V-LARGE-DOC '

Labels	Percentages
E	30
S	40
C	0
I	30

- 5) thenlper/gte-base: The GTE models are trained on a large-scale corpus of relevance text pairs, covering a wide range of domains and scenarios. This enables the GTE models to be applied to various downstream tasks of text embeddings, including information retrieval, semantic textual similarity, text reranking, etc. A lightweight mean pooling is applied to get the text representations: [20]

$$x = \frac{1}{n} \sum_{i=1}^n h_i \in \mathbb{R}^d \quad (2)$$

Following are the percentages of E, S, C, I labels for this particular query

C. Models for Autocompletion

To conduct the experiment, three models were leveraged to facilitate us with the auto-completion task:

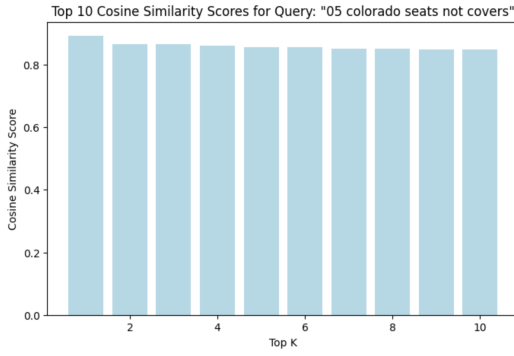


Fig. 7. Top 10 cosine similarity scores of a sample query

TABLE VII
PERCENTAGES OF E, S, C, I LABELS USING 'THENLPER/GTE-BAS'

Labels	Percentages
E	20
S	0
C	0
I	80

- 1) LSTM: The first implementation utilized a PyTorch based Long Short-Term Memory (LSTM) auto-completion model which predicts the next word in a sequence given a context of preceding words. The architecture consists of an embedding layer, LSTM layers to capture sequential dependencies, and a linear layer that predicts the probability distribution over the vocabulary. Adam optimizer and cross-entropy loss were used for parameter optimization. The model's performance is evaluated using metrics such as average loss, validation loss, and perplexity during the training process.

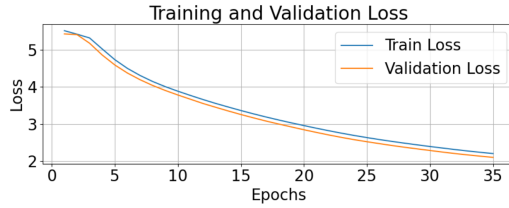


Fig. 8. Average and validation loss of LSTM

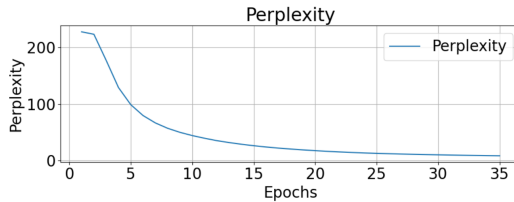


Fig. 9. Perplexity for LSTM

It was trained on a Wikipedia dataset, showing decreasing average and validation loss but high perplexity

(ranging approximately 490.82 to 469.28). When the model was tested, it produced sub-optimal outputs and indicated low accuracy.

- 2) Facebook/bart-large: The pre-trained BART (Bidirectional and Auto-Regressive Transformers) model is a neural network architecture developed by Facebook AI. BART employs both auto-regressive and bidirectional mechanisms during training, to effectively capture context and generate coherent and contextually relevant outputs. In inference, users can input a prompt or sentence with a placeholder (e.g., `mask`) for BART to complete. The model generates text based on the context provided, producing meaningful and contextually appropriate language. On average, the model achieves an overall accuracy of 66.87%.
- 3) N-gram: N-grams are contiguous sequences of n items from a given sample of text or speech. This approach creates and utilizes n-grams from a dataset of text queries. The n-gram model allows for predicting the next word based on the context of the preceding words, providing a foundation for word prediction, autocompletion, and language modeling tasks. This model was trained on the cleaned dataset generated after EDA, comprising 20,000 queries. To check the distribution and diversity of n-grams in the training data, the counts of n-grams were assessed. The relatively high counts indicated a diverse vocabulary and testing on dummy data showed promising results as displayed in Table 8.

TABLE VIII
N-GRAM RESULT

n-gram	count
Uni-gram	583
Bi-gram	1441
Tri-gram	2016
Quad-gram	2493
Quint-gram	2952

When this model was evaluated using dummy data similar to expected user queries, the model seemed to perform well, attributing highest probability to the most appropriate next word.

D. Spell correction

Multiple models such as 'spelling-correction-english-base' built upon the BART architecture, were tested on the data under consideration for this particular task. However, their results failed to meet the desired performance criteria. Consequently, instead of training a new model on this dataset, the existing Upstart Commerce API for spell correction was procured to ensure better efficiency of results. This API was already fine tuned to meet the requirements posed by the collaborator Upstart Commerce.

E. Finetuning Semantic Search

The figure clearly elaborating the procedure of fine tuning has been appended as figure 10, please reference it for the

following section.

The diagram has been made using MiniLM as the example model that is being used, any other relevant model can be replaced here as well. Consequently some parts for elaboration of the system have been taken from Wang et al. [15] [21] and Vaswani et al. [22]. In the fine-tuning process of MiniLM, the model is utilized for embedding generation through feed-forward operations when presented with queries and products. Following this, cosine similarity calculations are performed on the embeddings derived from the query and product. Subsequently, the model is instructed to generate the top 100 matching results using the rank@100 function. A comparative analysis is then conducted between the obtained top 100 results and the expected outcomes. The information gleaned from this comparison is systematically back propagated to update the model's layers, thereby refining its parameters based on the evaluation of actual versus expected results. This iterative process contributes to the ongoing optimization of MiniLM for enhanced performance in specific language-related tasks. Refer to figure 13 under the experimental results section to see the comparison of all-MiniLM-L6-v2 model before and after fine-tuning.

V. EXPERIMENTAL RESULTS

This section will define how the dataset was divided, the experimental setup within which all experiments were performed, the basis upon which the models were evaluated and the results attained.

A. Dataset Division

The data followed an 80-20 split where 80% of the data was used for training the models and 20% was used for testing. This was from the approximately 20,000 cleaned, pre-processed, Upstart-Amazon merged queries data that had been extracted for this task.

B. Experimental Setup

The model undergoes training within the Tensorflow:2.11 environment leveraging the capabilities of Google Cloud Platform's Vertex AI. The machine configuration selected for this task includes 60GB RAM, 16 vCPUs and a dedicated T4 x 1 NVIDIA GPU.

C. Evaluation Criteria

The evaluation criteria for semantic searching and other models is as follows below;

1) *Semantic Searching Evaluation Criteria:* The models are evaluated on the basis of how well they perform semantically, for this task it is imperative that they return accurate results for queries. Many different methodologies that can be used to evaluate the models once they have generated a list of semantically matching results have been read when it comes to evaluation metrics. Our main aim while evaluating these models is to see their:

- Semantic understanding capabilities.
- Ability to handle contextual information.

- Efficiency in processing queries and documents.
- Compatibility with our dataset.

Eventually the evaluation metrics selected for this task were nDCG (Normalized Discounted Cumulative Gain), MRR (Mean Reciprocal Rank) and MAP (Mean Average Precision). The formula for nDCG is:

$$\text{nDCG@k} = \frac{\text{DCG@k}}{\text{IDCG@k}} \quad (3)$$

Where DCG@k (Discounted Cumulative Gain at position k) is calculated as:

$$\text{DCG@k} = \sum_{i=1}^k \frac{2^{\text{rel}_i} - 1}{\log_2(i + 1)} \quad (4)$$

IDCG@k (Ideal Discounted Cumulative Gain at position k) is:

$$\text{IDCG@k} = \sum_{i=1}^k \frac{2^{\text{rel}_i} - 1}{\log_2(i + 1)} \quad (5)$$

In both formulas: rel_i is the relevance of the item at position i in the ranking list. k is the position cutoff for the nDCG calculation. Otherwise the top k results.

The formula for MRR is:

$$\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i} \quad (6)$$

Where $|Q|$ is the total number of queries evaluated. rank_i is the rank of the first relevant item for the query at position i .

The formula for MAP is:

$$\text{MAP} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \text{AP}_i \quad (7)$$

Where $|Q|$ is the total number of queries being evaluated. AP_i is the average precision for the query at position i .

Average Precision (AP) for a single query is calculated as follows:

$$\text{AP}_i = \frac{1}{\text{num_relevant}_i} \sum_{k=1}^N P(k) \times \text{rel_at_k}_i \times \text{is_relevant}_k \quad (8)$$

Where num_relevant_i at i for the query is the total number of relevant items. $P(k)$ is the precision at cutoff k (the number of relevant items retrieved so far). rel_at_k_i is a binary indicator function that denotes whether the item at position k is relevant to the query at position i . is_relevant_k is a binary indicator function that denotes whether the item at position k is relevant, with a value of 1 if relevant and 0 otherwise.

2) *Autocomplete and Spell Correction Evaluation Criteria:* For Spell correction, on the list of specific words provided by Upstart Commerce, models were evaluated. The basis of evaluation was whether the models managed to return the word expected in the list. Accordingly for each correct result a counter got incremented and eventually was divided by the total number of words on the list to attain the accuracy of the model for this specific data.

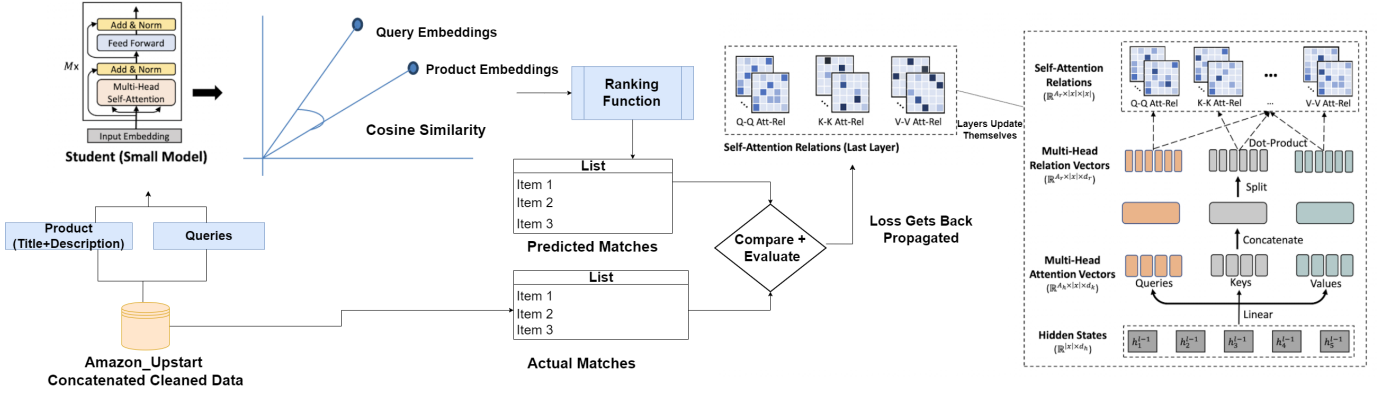


Fig. 10. Diagram depicting the entire process flow of the data training/fine tuning.

For Autocomplete models, evaluation was done on the basis of their accuracy, perplexity, recall, precision and F1 scores. Alongside this their average performance times were also taken under consideration.

The formula for Perplexity is:

$$\text{Perplexity} = e^{\frac{1}{N} \sum_{i=1}^N -\log(p(x_i))} \quad (9)$$

Precision is calculated using the formula:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (10)$$

Recall has the formula:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (11)$$

F1 score calculation utilized both precision and recall in the formula:

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (12)$$

D. Results

The results obtained by the semantic searching models before fine tuning are mentioned in the table below:

TABLE IX
PERFORMANCE METRICS

Model	nDCG	MRR	MAP
all-MiniLM-L6-v2	0.9301	0.77	0.0057
msmarco-distilroberta-base-v2	0.9274	0.75	0.0044
Msmarco-distilbert-base-v3	0.9331	0.75	0.0049
naver/efficient-splade-V-large-doc	0.9311	0.80	0.0036
thenlper/gte-base	0.9381	0.76	0.0068

These results are depicted in the Fig. 11 and 12.

As can be seen, the results of these models were pretty competitive. However after fine tuning the best results acquired were from the All-MiniLM-L6-v2 model. Consequently it was chosen as the best model.

Fig. 13 demonstrates MiniLM's results against an example query from the dataset.

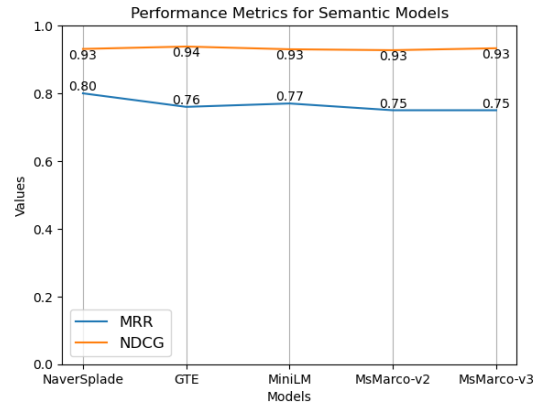


Fig. 11. Model Results for Evaluation Metrics 1

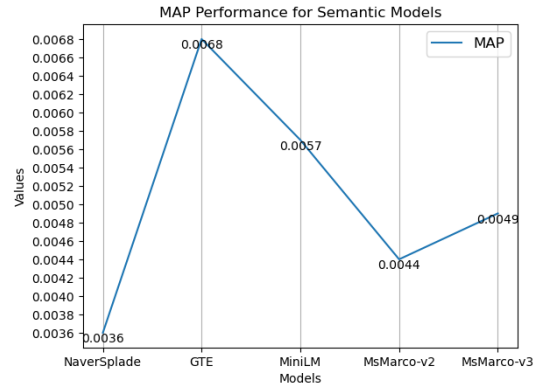


Fig. 12. Model Results for Evaluation Metrics 2

VI. CONCLUSION

In this paper we compare deep learning-based semantic search techniques to enhance product matching in e-commerce by evaluating their performance on metrics such as nDCG, MRR and MAP scores. Deep learning techniques are also used for implementing spell correction and autocomplete of queries. The amazon shopping queries data set served as our primary source of training and evaluation material alongside

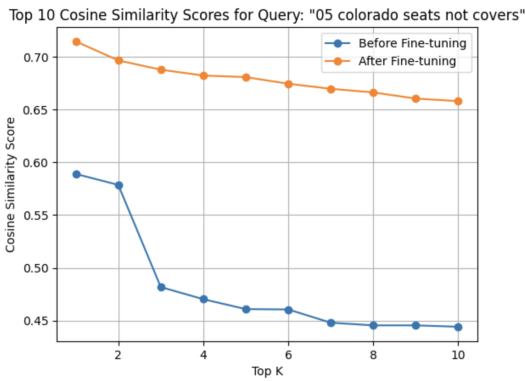


Fig. 13. Fine-tuning result of 'all-MiniLM-L6-v2'

the Upstart Catalog. For semantic searching various models such as naverSPLADE, MiniLM, distilBERT, distilRoBERTA and GTE-base were fine tuned and compared to one another. The best performing for our scenario, MiniLM was then chosen and deployed on OpenSearch. Future works can improve upon the existing ground work by attempting additional techniques for semantic enrichment of queries as well as tailor search results for the user based on the user's recent product search history.

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